Examiner only

6. *(a)* Nuclear fusion and nuclear fission reactions both produce heat energy. Describe and compare controlled nuclear fission and nuclear fusion reactions. [6 QER]

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(b) A typical fusion reaction between tritium and deuterium is:

$$\text{tritium} + \text{deuterium} \longrightarrow \text{helium} + \text{neutron} + 2.82 \times 10^{-12}\,\text{J of energy}$$

Calculate the number of fusion reactions needed each second to provide an energy output of 6.2 MJ every second. [2]

Number of reactions = ...

8